

CHAPTER 13

LINEAR MODELS

13.1 INTRODUCTION

This chapter is devoted to statistical problems arising in situations where the observed values (called “dependent variable,” “response,” etc.) are influenced by some other variable (referred to as “independent variable,” “explanatory variable,” “treatment,” “factor,” etc.). In this chapter, we consider the case where only expected values are affected, while other characteristics (as well as the type of distribution) remain the same.

The theories that we present depend on the assumptions about the independent variable: Can its value be observed or not? If yes, can it be controlled by the experimenter? If no, can it be regarded as random? Depending on the answer to these questions, we have a regression analysis or different analysis of variance (ANOVA) models (one-factor, two-factor, with or without interaction, with fixed or random effects, etc.).

Many models (theories) are available in the general setup considered in this chapter. The basic ideas and solutions we present cover a few of the most representative cases. We hope that the information we provide is enough information to motivate users of various statistical packages to try to identify conditions where a specific procedure is not the only one available. A thorough and exhaustive presentation of any of these theories can be found in any of numerous books devoted to regression analysis or ANOVA (e.g., see Montgomery et al., 2006; Stapleton, 1995; Myers and Milton, 1991; Myers, 1986).

In this chapter, we outline some of the most common ways of analysis of data measured on a scale of at least interval type. We discuss the methods of detecting and measuring effects expressed through the mean of the observed random variable. However, we present only the most common of such methods, regression analysis and ANOVA.

■ EXAMPLE 13.1

In a simple case of linear regression analysis, we observe a random variable Y such that $Y = \alpha X + \beta + \xi$, where X is some variable (random or not, possibly controlled

by the experimenter) and ξ is the “error,” meaning a random variable with $E(\xi) = 0$, $\text{Var}(\xi) = \sigma^2 < \infty$. Both X and Y are assumed to be measured on an interval (or possibly even ratio) scale.

Specific examples may be obtained by taking Y to be the time to completion of some chemical reaction, and X being the temperature; Y may be some substance that accumulates linearly (up to random fluctuations) in human bones throughout life, and X may be the age of the person at death, and so on.

Typically, we have data in the form of pairs (X_i, Y_i) , $i = 1, \dots, n$, possibly with some X_i 's repeating. The problems are to estimate α, β , and σ^2 (or to test hypotheses about these parameters) and predict the value of Y to be observed for some X_0 , or to estimate X corresponding to some observed value of Y .

Obvious generalizations involve a model with more than one variable X , for instance, when $Y = \alpha_1 X_1 + \alpha_2 X_2 + \beta + \xi$, where X_1 and X_2 are some variables. We may have nonlinear models, such as $Y = \alpha X^2 + \beta X + \gamma + \xi$ (quadratic regression), etc.

■ EXAMPLE 13.2 Analysis of Variance or ANOVA

ANOVA applies to situations where variables X_i are nominal. We typically speak of “factors” that operate on some levels.³⁰ For instance, we might have data (measurements of some “response”) taken from populations classified according to some criteria such as sex (male and female) and smoking status (never smoked, former smoker, and current smoker). We want to find out whether any of these factors (sex and smoking status) has an effect on Y (e.g., response to some drugs).

The data can be summarized in a table:

		1	2	3
		Never smoked	Former smoker	Current smoker
1	Male	Y_{11}	Y_{12}	Y_{13}
2	Female	Y_{21}	Y_{22}	Y_{23}

Here Y_{ij} is the response of a subject from group (i, j) , where we have one observation in each cell. The model is

$$Y_{ij} = \mu + \alpha_i + \beta_j + \xi_{ij},$$

for which we can assume $\alpha_1 + \alpha_2 = 0$, $\beta_1 + \beta_2 + \beta_3 = 0$, and ξ_{ij} to be the error term satisfying $\xi_{ij} \sim N(0, \sigma^2)$. We may wish to test the hypotheses $H_0 : \alpha_1 = \alpha_2 = 0$ and $H'_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (no sex effect, and no smoking effect) against the alternatives $H_1 : \alpha_1 \neq \alpha_2$ and $H'_1 : \beta_k \neq \beta_l$ for some $k, l = 1, 2, 3, k \neq l$.

The scheme of Example 13.2 can be modified in a number of ways, of which we will discuss some. The main issue we note here is that comparing the results within each pair separately (until either a pair of “significantly different” results is found or until all pairs are checked and found not to differ significantly) is *not a correct method*. Indeed, for many pairs the probability of finding a pair with a large difference becomes quite likely because of chance fluctuations, even where the null hypothesis is in fact true. The correct method requires testing all pairs *at once*, which is accomplished by the ANOVA methodology.

³⁰The use of the word “level” in the context of ANOVA does not imply any specific ordering of the levels of factors such as “sex,” F (female) and M (male).

13.2 REGRESSION OF THE FIRST AND SECOND KIND

Let Y denote a dependent variable (assumed to be one-dimensional) and X (or $\mathbf{X}$) denote an independent variable. In most cases X will be one-dimensional, but the considerations can usually be extended to m -dimensional vectors $\mathbf{X} = (X_1, \dots, X_m)$, where $m > 1$.

A few comments about the nature of $\mathbf{X}$ and some illustrative examples are in order here. We start with the case where the values of the independent variable (or variables) can be observed. Depending on how the values of $\mathbf{X}$ are chosen, $\mathbf{X}$ may be random, either one- or multidimensional, nonrandom, or even under the experimenter's control. Typical cases of such situations are exemplified below.

■ EXAMPLE 13.3

In some genetic theories one studies a characteristic Y of offspring, such as height, which depends on characteristics $\mathbf{X} = (X_1, X_2)$ of a father and a mother (X_1, X_2 may be heights, but it is also possible to study some other features affecting the offspring's height Y).

■ EXAMPLE 13.4

Sometimes the randomness of Y for a given $X = x$ has to be postulated, since it is not possible to observe more than one Y for the same x . This happens, for instance, if x is the calendar time, and we observe the stock market index Y_x at the end of the day x . Here we treat Y_x as random, since we cannot predict its value with complete precision. We only have the conviction that "if such and such events had occurred, the value Y_x would be different from that being observed." But for each x only one single value is recorded, and we have no empirical access to the distribution unless we make some assumptions about the nature of randomness of Y_x across different times x .

■ EXAMPLE 13.5

It may happen that X is not random. For instance (in the case of one dimension), we may study the relation between some developmental characteristic Y of a child, say height or size of vocabulary, as dependent on age X . The randomness concerns values of Y for the same $X = x$, since in the population of children of age x there is some variability of values of Y .

■ EXAMPLE 13.6

It may even happen that the values of X are totally under the experimenter's control. For instance, a chemist may be interested in some characteristic Y of a chemical reaction (e.g., its duration) depending on temperature X . Then X can be determined arbitrarily by the experimenter, and for a given temperature $X = x$, the randomness of Y may be due to measurement error, or to some other factors.

In each of the cases under consideration, the data have the form of a set of pairs (x_i, y_i) , $i = 1, 2, \dots, n$, where values of x_i can repeat. Even if the x_i 's are not random, the formulas are identical with those obtained under the assumption that X has discrete uniform distribution over the set $\{x_1, \dots, x_n\}$, with probabilities appropriately increased in the case of repeated values. Thus, we will proceed as if X (respectively, $\mathbf{X}$) were a random variable. We will assume throughout this chapter that X and Y have finite variances, hence also finite expectations.

Let us begin by recalling some facts from Chapter 7. Suppose that we know the value of X , $X = x$, and we want to predict the value of Y . The best prediction of Y (in the sense of mean square error) is given by the conditional expectation

$$u(x) = E(Y|X = x). \quad (13.1)$$

In other words, we have

$$\min_{\xi} E[(Y - \xi)^2|X = x] = E[(Y - u(x))^2|X = x].$$

Accordingly, we introduce the following definition:

Definition 13.2.1 The conditional expectation of Y given $X = x$, that is, the function $u(x)$ given by (13.1), will be called the *true regression*, or regression of the *first kind*, of Y on X . $\square$

In the case of a continuous random variable X , it may happen that the regression function $u(x)$ is defined only for almost all x , that is, on a set A such that $P\{X \in A\} = 1$. However, in all cases that are encountered in statistics, there is typically a function $u(x)$ that is “natural” in a given problem. We will select such a function and comment on the nonuniqueness of a regression function only when such a comment is essential for the problem.

The usefulness of the regression function $u(x)$ is not restricted to the prediction of Y . To mention just one other use of it, suppose that we do not know how X and Y are related; we only formulated the hypothesis that the true regression function is equal to some function u_0 . In other words, the null hypothesis is $H_0 : E(Y|X = x) = u_0(x)$ for all x . If we know something about the conditional distribution of Y given X , we may test this hypothesis, rejecting it if the value of Y observed for given $X = x$ is “far” from $u_0(x)$.

Finding the true regression $u(x)$ requires the knowledge of the joint distribution of (X, Y) or, more precisely, the conditional distribution of Y given $X = x$. Such knowledge can come from a sufficiently deep understanding of the stochastic mechanisms that connect X and Y , or from very extensive data on pairs (X_i, Y_i) . There are many practical cases when neither is available. Moreover, even if we do know the joint distribution of (X, Y) , determining $u(x)$ may present formidable difficulties.

To cover such situations, another type of regression has been introduced. Starting again from problem of prediction, suppose that we want to find the best predictor of Y that is linear in X . It will have the form $Y_p = a + bX$, and we will have to find such a and b that Y_p is the best predictor of Y which will minimize the mean square error $E(Y - Y_p)^2$. We have here

$$\begin{aligned} E(Y - Y_p)^2 &= E[Y - (a + bX)]^2 \\ &= E(Y^2 + a^2 + b^2X^2 - 2aY - 2bXY + 2abX) \\ &= E(Y^2) + a^2 + b^2E(X^2) - 2aE(Y) - 2bE(XY) + 2abE(X). \end{aligned}$$

By differentiating with respect to a and b and setting the derivatives to be 0, we obtain the *normal equations*:

$$\begin{aligned} a + bE(X) &= E(Y), \\ aE(X) + bE(X^2) &= E(XY). \end{aligned}$$

Multiplying the first equation by $E(X)$ and subtracting it from the second, we obtain

$$b = \frac{E(XY) - E(X)E(Y)}{\sigma_X^2} = \rho \frac{\sigma_Y}{\sigma_X} \quad (13.2)$$

and consequently

$$a = E(Y) - \rho \frac{\sigma_Y}{\sigma_X} E(X). \quad (13.3)$$

Thus coefficients a and b require only means, variances, and covariances of X, Y . We now introduce the following definition:

Definition 13.2.2 The relation

$$Y = a + bX = E(Y) + \rho \frac{\sigma_Y}{\sigma_X} [X - E(X)] \quad (13.4)$$

is called the *linear regression*, or regression of the *second kind*, of Y on X . □

■ **EXAMPLE 13.7**

Let us consider the situation where the true regression is the function

$$u(x) = \begin{cases} x & \text{for } 0 \leq x \leq \frac{1}{2} \\ \frac{1}{2} & \text{for } \frac{1}{2} \leq x \leq 1. \end{cases}$$

Such a regression can occur for various conditional distributions of Y . To fix the idea, suppose that given x , the random variable Y is distributed uniformly on an interval whose leftmost point is 0. Thus, for $0 \leq x \leq \frac{1}{2}$, random variable Y_x is uniform on $[0, 2x]$, whereas for $\frac{1}{2} \leq x \leq 1$, random variable Y_x is uniform on $[0, 1]$. The situation is presented in Figure 13.1. The shaded area shows all possible points (X, Y) . The polygonal line OAB is the graph of the true regression $u(x)$. For use in the next example, note that we have here

$$\text{Var}(Y|X = x) = \begin{cases} \frac{x^2}{3} & \text{for } 0 \leq x \leq \frac{1}{2} \\ \frac{1}{12} & \text{for } \frac{1}{2} \leq x \leq 1. \end{cases} \quad (13.5)$$

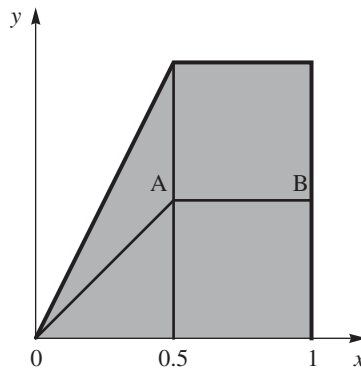


Figure 13.1 True regression.

■ **EXAMPLE 13.8**

In Example 13.7, the distribution of X did not play a role in determining regression. In fact X could have been nonrandom. On the other hand, to determine the linear regression of Y on X , it is necessary to know the marginal distribution of X . The intuitive

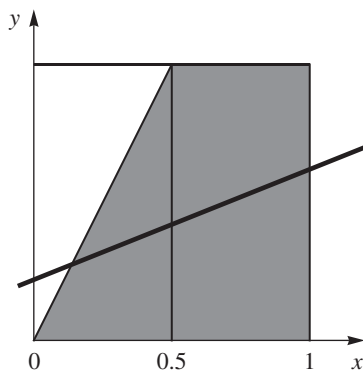


Figure 13.2 Linear regression for uniform distribution of (X, Y) .

justification here is that in linear regression the objective is to find the best approximation of Y by a straight line. Thus, it matters which values of X occur more often and which occur less often. Suppose that the joint distribution of the vector (X, Y) is uniform on the shaded area in Figure 13.2. To determine the linear regression, we need the expectations and variances of X and Y as well as their covariance. We have here, using simple geometry, for the marginals of X and Y :

$$f_X(x) = \begin{cases} cx & \text{for } 0 \leq x \leq \frac{1}{2} \\ \frac{c}{2} & \text{for } \frac{1}{2} \leq x \leq 1. \end{cases} \quad (13.6)$$

The value of c is determined from the condition

$$1 = \int_0^1 f_X(x) dx = \frac{3}{8}c;$$

hence $c = 8/3$. Consequently,

$$E(X) = \int_0^1 x f_X(x) dx = \frac{11}{18}.$$

After simple integration we obtain $E(X^2) = 31/72$, hence $\text{Var}(X) = 37/648$. Similarly, $f_Y(y) = (4/3)(1 - y/2)$ for $0 \leq y \leq 1$, and consequently $E(Y) = 4/9$, and $\text{Var}(Y) = 13/162$.

Finally, the joint density $f(x, y)$ is constant, equal to $4/3$ on the shaded area in Figure 13.2. Thus, $E(XY) = 7/24$, and using (13.4), we obtain the equation for linear regression (see Figure 13.2):

$$Y = \frac{X}{4} + \frac{7}{24}.$$

It is of some interest to compare the average square error of prediction of Y on the basis of X if we use true and linear regressions. For prediction under true regression, we find the average square error, using (13.5) and (13.6):

$$\begin{aligned} E_X[\text{Var}(Y|X)] &= \int_0^1 \text{Var}(Y|X = x) f_X(x) dx \\ &= \int_0^{1/2} \frac{x^2}{3} \times \frac{8}{3} x dx + \int_{1/2}^1 \frac{1}{12} \times \frac{4}{3} x dx = \frac{5}{72} = 0.0694. \end{aligned}$$

For prediction with the use of linear regression, we have the error

$$\begin{aligned} E\left(Y - \frac{X}{4} - \frac{7}{24}\right)^2 &= E(Y^2) + \frac{1}{16}E(X^2) + \left(\frac{7}{24}\right)^2 - \frac{1}{2}E(XY) \\ &\quad - \frac{7}{12}E(Y) + \frac{1}{48}E(X) = 0.0738. \end{aligned}$$

Thus, in this case, the mean square error of the linear prediction is about 5.5% higher than the corresponding error of prediction based on the true regression. Whether a 5.5% difference is important or negligible depends on the context. The point is that the mean square error of a linear prediction is *always* at least as large as the error of prediction based on a true regression. Also one can easily construct examples where the ratio of these two errors is as large as one wishes. In the extreme case, if $Y = g(X)$, where g is a deterministic nonlinear function, g is the true regression and there is no error involved in a prediction of Y using $g(X)$ as the predictor. But there is an error involved if a nonlinear function g is approximated by a linear function $a + bX$.

Example 13.8 shows that the issue of prediction errors in comparison of two types of regression is only partially statistical. One component is due to the randomness of Y for a given $X = x$, and the other is due to replacing the true regression $u(x)$ by a straight line.

The situation may be improved by considering a regression of the second order, where the predictor is of the form $Y_p = a + bX + cX^2$, or a regression of some other special form, and as $Y_p = a + b \sin(cX)$, and so on. The point is that none of these regressions can give the mean square error lower than that based on true regression.

PROBLEMS

- 13.2.1** Let X and Y have a joint distribution uniform on a parallelogram with vertices at points $(-1, -1)$, $(0, -1)$, $(1, 1)$, and $(0, 1)$. Find the true and linear regression of: (i) X on Y . (ii) Y on X .
- 13.2.2** Find an example of random variables (X, Y) such that their true and linear regressions of Y on X coincide, but (X, Y) do not have a joint normal distribution.
- 13.2.3** Suppose that the true regression of Y on X is not linear in X . Is it possible that the marginal distribution of X is such that the expected square error of the best linear predictor of Y is the same as the expected square error of predictor based on the true regression u ?
- 13.2.4** Let (x_i, y_j) , $i = 1, \dots, n$ be the data where at least one value of x_i is not zero. Find an estimate of the slope parameter b if it is known that $a = 0$ [i.e., find the best fit of the model $E(Y|x) = bx$].
- 13.2.5** Suppose that $\mathbf{X} = (X_1, X_2)$. Find coefficients a, b_1, b_2 such that $Y_p = a + b_1X_1 + b_2X_2$ is the best linear predictor of Y given $\mathbf{X}$.
- 13.2.6** Find true regression Y on X if X and Y have a joint trinomial distribution

$$P\{X = x, Y = y\} = \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y (1-p_1-p_2)^{n-x-y}$$

for $x = 0, 1, \dots, n, y = 0, 1, \dots, n$, and $0 \leq x + y \leq n$.

- 13.2.7** The number of eggs in nests of a certain species of birds is one, two, or three, with two eggs found in about 80% of nests, and one or three eggs in about 10% of nests

each. In one-egg nests, the egg hatches successfully in 75% of cases. In two-egg nests, the probabilities for the number of offspring are 0–20%, 1–30%, 2–50%, whereas in three-egg nests, these probabilities are 0–10%, 1–20%, 2–60%, 3–10%. Find the best linear predictor of Y being the number of offspring based on the observation of X —the number of eggs in the nest.

13.3 DISTRIBUTIONAL ASSUMPTIONS

In addition to the classification of regression models with respect to the two types of regression, a meaningful and useful classification is obtained when one considers typical assumptions about the distribution of the response Y for a given value of X .

The oldest methods that can be included in regression theory dates back to the beginning of the nineteenth century and the works of Legendre. He found a method of best approximation of a set of points by a straight line. The word “best” was understood in the sense of least squares. In other words, Legendre found, for a set of data points (x_i, y_i) , the coefficients a and b such that the sum $\sum_i [y_i - (a + bx_i)]^2$ attained its minimum. In this formulation, no assumptions about randomness are needed: the best-fitting line always exists, even if the “best” fit does not mean a “good” fit. The extension to the best linear fit in two (or k) dimensions—finding a_1, a_2, b to minimize the sum $\sum [y_i - (a + b_1 x_i^{(1)} + b_2 x_i^{(2)})]^2$ —is now straightforward. Similarly, the theory extends naturally to other forms of relations, such as quadratic or periodic functions obtained by minimizing the sums $\sum_i [y_i - (a + bx_i + cx_i^2)]^2$ or $\sum_i [y_i - a \cos(bx_i + c)]^2$.

Viewed in this way, regression theory belongs properly to the domain of numerical analysis. However, to allow statistical inference, one usually makes some assumptions about the randomness inherent in the model. One of the standard assumptions is that for every x we have $Y = \varphi(x) + \epsilon$, where φ is some deterministic function and ϵ is the “error” random variable, such that:

- (i) $E(\epsilon) = 0$.
- (ii) $\text{Var}(\epsilon) = \sigma^2 > 0$.
- (iii) The errors $\epsilon, \epsilon', \epsilon'', \dots$, corresponding to different observations of Y (for the same, or for distinct values of x are uncorrelated.³¹

Under these assumptions, $\varphi(x)$ is the true regression of Y on X . If one now imposes some parametric model on φ , one can set up the least square equations for parameters of φ and the parameter σ^2 .

If the assumptions (i)–(ii) are replaced by

- (iv) The errors $\epsilon, \epsilon', \dots$, have a $N(0, \sigma^2)$ distribution,

then the errors are independent. The independence of errors and knowledge of the distribution, have profound theoretical consequences: we can write the likelihood of the data and find the maximum likelihood estimators of the parameters.

One can easily see that in this case the maximum likelihood estimate (MLE’s) coincide with least square estimators, so it might appear that nothing is gained by replacing (i)–(iii) with (iii) and (iv). However, under (iv) we know the sampling distribution of the estimators, and we are therefore able to use the results from estimation and testing theory to build confidence intervals, tests of various hypotheses, and so on.

³¹The assumption that variance is independent of x is often called homoscedasticity, as opposed to *heteroscedasticity*.

Finally, let us mention, that if in addition to (i), (ii), and (iv) we can assume that

(v) $\varphi(x) = a + bx$,

(vi) X has a normal distribution,

then the joint distribution of (X, Y) is bivariate normal. The converse is also true, as shown in Theorem 8.5.4. We may summarize these facts as the following:

Theorem 13.3.1 *In case of bivariate normal distribution (X, Y) , the true and linear regression coincide.*

A comment appears necessary here in regards to assumptions (iii) and (iv), and their implementation in practical situations. To illustrate potential difficulties, suppose that we collect data on a regression where x is the age of a child, and Y is some response, such as reaction time to a specific stimulus. Suppose that we need two observations for the same age x . Then it is not correct to measure the reaction time Y twice for the same child of age x , even if such observations can be regarded as independent. The correct procedure is to take two children of the same age x and observe their reaction times. The reason is that variability of Y has two components: the between-children variability and the within-child variability. Taking two measurements for the same child will involve only the second component, whereas other observations will involve both components, violating (among others) the assumption of homoscedasticity.

Assumptions (i)–(vi) are by no means the only sensible assumptions one can make in regression analysis. The linearity assumption $E(Y_x) = a + bx$ is often acceptable as an approximation or in some narrow range. A typical case of this kind is illustrated by the following example:

■ EXAMPLE 13.9 Logistic Regression

Assume that the response variable Y_x is of the binary character (e.g., success or failure). We can always take the possible values of Y_x as 0 and 1 so that $E(Y_x) = P\{Y_x = 1\} = \pi(x)$. We will assume that x is a numerical variable (random or not, depending on the situation under study), and that $\pi(x)$ is the true regression of Y on x . The assumption $\pi(x) = a + bx$ can be realistic only in a narrow range of values of x , since we must have $0 \leq \pi(x) \leq 1$ for all x . Still the inference about the shape of function $\pi(x)$ is of considerable interest, and linearity is a powerful assumption (in the sense of allowing many analytical results). To realistically utilize such an assumption, we consider the odds ratio $\pi(x)/[1 - \pi(x)]$, with range $(0, \infty)$, and its logarithm, with range $(-\infty, \infty)$. Thus, we consider the model

$$\log \frac{\pi(x)}{1 - \pi(x)} = a + bx,$$

or equivalently,

$$\pi(x) = \frac{e^{a+bx}}{1 + e^{a+bx}}. \quad (13.7)$$

The right-hand side of (13.7) is called the *logistic function* (and is a cdf of a logistic distribution if $b > 0$), which explains the name logistic regression. The problem now is to estimate parameters a and b given the data (x_i, n_i, N_i) , $i = 1, \dots, m$, where x_i 's are the values of the independent variable at which observations are taken, while n_i and N_i are the number of successes and the number of trials at value x_i , respectively. The

likelihood is therefore

$$\begin{aligned} L &= \prod_{i=1}^m \binom{N_i}{n_i} [\pi(x_i)]^{n_i} [1 - \pi(x_i)]^{N_i - n_i} \\ &= \left[\prod_{i=1}^m \binom{N_i}{n_i} \right] \frac{e^{a \sum n_i + b \sum x_i n_i}}{\prod_{i=1}^m (1 + e^{a + bx_i})^{N_i}}. \end{aligned}$$

Determining a and b that maximize L requires using numerical iterations.

PROBLEMS

13.3.1 The random variable Y_x has the density

$$f(y|x) = \begin{cases} 1 & \text{for } a + bx - \frac{1}{2} < y < a + bx + \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$

Find the MLE's of a and b given the sample: $(-1, 1.3), (0, 1.4), (1, 0.1), (2, -0.4), (3, -0.1)$. Compare the MLE's with the LS estimates.

13.3.2 Let (X, Y) have the distribution uniform on the quadrangle with vertices $(0, 0), (1, 1), (0, 2)$, and $(0.5, 1)$. Find: **(i)** The true regression of Y on X and of X on Y . **(ii)** $\text{Var}(X|Y)$ and $\text{Var}(Y|X)$.

13.3.3 Assume that the total number X of eggs laid by a bird has a Poisson distribution with parameter λ , and that each egg hatches with probability p , independent of other eggs. Find the regression of the number Y of eggs hatched on X . Let X_i, Y_i be the number of eggs laid and hatched by the i th bird. Derive the likelihood of the data for n birds, and the equation for estimating λ and p . Find the MLE's of λ and p .

13.3.4 Answer Problem 13.3.3 assuming a more realistic model, where X_i 's have a Poisson distribution conditional on a positive value, that is, for $k = 1, 2, \dots$,

$$P\{X_i = k\} = \frac{(\lambda^k/k!)e^{-\lambda}}{1 - e^{-\lambda}}.$$

13.4 LINEAR REGRESSION IN THE NORMAL CASE

In this section, we will present the main results in the case where for any given x , the response Y is of the form

$$Y = a + bx + \epsilon$$

for $\epsilon \sim N(0, \sigma^2)$. Moreover, we will assume that the errors ϵ for different observations of Y (for the same as well as for a distinct x) are independent.

We will assume that the data have the form

$$(x_i, y_i), \quad i = 1, \dots, n, \tag{13.8}$$

where $y_1, \dots, y_n$ are independent observations. We will use the notation

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

and assume that not all x_i 's are identical.

The estimates of parameters a, b , and σ^2 are easy to obtain. The likelihood function of the sample (13.8) is

$$L = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-(y_i - a - bx_i)^2 / 2\sigma^2}$$

$$= \sigma^{-n} (2\pi)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - a - bx_i)^2 \right\};$$

hence

$$\log L = C - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - a - bx_i)^2.$$

Differentiating with respect to a and b , we obtain a pair of equations (in which σ^2 cancels out) that are the same as the equations for the least squares estimates of a and b . The solutions can be obtained from formulas (13.2) and (13.3) by treating X as a random variable uniformly distributed on $x_1, \dots, x_n$. Thus

$$\hat{b} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad \text{and} \quad \hat{a} = \bar{y} - \hat{b}\bar{x}. \quad (13.9)$$

The denominator in the expression for $\hat{b}$ is not zero in view of the assumption that not all x_i 's are equal.

Differentiating $\log L$ with respect to σ^2 and setting the derivative equal to 0, we obtain the MLE of σ^2 :

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{a} - \hat{b}x_i)^2. \quad (13.10)$$

To determine the sampling distribution of estimators (under variability of y_j 's only, for fixed x_j 's), let us regard y_j as a value of the random variable Y_{x_j} . Then the estimators of b, a , and σ^2 are, respectively,

$$U = \frac{\sum (x_i - \bar{x})(Y_{x_i} - \bar{Y})}{\sum (x_i - \bar{x})^2}$$

$$V = \bar{Y} - U\bar{x}$$

$$T^2 = \frac{1}{n} \sum_{i=1}^n (Y_{x_i} - V - Ux_i)^2. \quad (13.11)$$

Notice that the randomness of U, V , and T^2 is related only to the variability of the dependent variable Y_x about its mean $a + bx$. The values x_i may be nonrandom, and even if they are arising from the sampling, U, V , and T^2 involve only a conditional distribution of Y given $X = x$.

It follows from (13.11) that U , as a linear combination of normally distributed random variables, has itself a normal distribution. To determine the mean, observe that

$$E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(Y_{x_i}) = \frac{1}{n} \sum_{i=1}^n (a + bx_i) = a + b\bar{x}.$$

Consequently, we have

$$E(U) = \frac{\sum (x_i - \bar{x})E(Y_{x_i} - \bar{Y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x})[a + bx_i - (a + b\bar{x})]}{\sum (x_i - \bar{x})^2} = b.$$

To find the variance of U , we write

$$U = \frac{\sum (x_i - \bar{x}) Y_{x_i} - \bar{Y} \sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x}) Y_{x_i}}{\sum (x_i - \bar{x})^2}.$$

Consequently, by assumption of homoscedasticity,

$$\sigma_U^2 = \text{Var} \left\{ \frac{\sum (x_i - \bar{x}) Y_{x_i}}{\sum (x_i - \bar{x})^2} \right\} = \frac{\sum (x_i - \bar{x})^2 \text{Var}(Y_{x_i})}{[\sum (x_i - \bar{x})^2]^2} = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}. \quad (13.12)$$

Next the estimator V given by (13.11) is also a linear combination of the normal random variables, so it has a normal distribution. Here

$$E(V) = E(\bar{Y}) - E(U)\bar{x} = a + b\bar{x} - b\bar{x} = a$$

so that U and V are unbiased estimators of b and a .

The derivation of the variance of estimator V is somewhat messy. We have

$$\begin{aligned} \sigma_V^2 &= \text{Var}(\bar{Y}) + \bar{x}^2 \text{Var}(U) - 2\bar{x} \text{Cov}(U, \bar{Y}) \\ &= \text{Var} \left(\frac{1}{n} \sum Y_{x_i} \right) + \bar{x}^2 \sigma_U^2 - 2\bar{x} \text{Cov} \left[\frac{\sum (x_i - \bar{x}) Y_{x_i}}{\sum (x_i - \bar{x})^2}, \frac{1}{n} \sum Y_{x_j} \right] \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{\sum (x_i - \bar{x})^2} - \frac{2\bar{x}}{n \sum (x_i - \bar{x})^2} \text{Cov} \left[\sum (x_i - \bar{x}) Y_{x_i}, \sum Y_{x_j} \right]. \end{aligned}$$

In the last term, all covariances corresponding to $i \neq j$ vanish while all others are equal σ^2 . Consequently

$$\text{Cov} \left[\sum (x_i - \bar{x}) Y_{x_i}, \sum Y_{x_j} \right] = \sum (x_i - \bar{x}) \sigma^2 = 0.$$

After some algebra we obtain

$$\sigma_V^2 = \sigma^2 \frac{\sum x_i^2}{\sum (x_i - \bar{x})^2}. \quad (13.13)$$

In a similar manner, we can show that

$$\text{Cov}(U, V) = -\sigma^2 \frac{\bar{x}}{\sum (x_i - \bar{x})^2} \quad (13.14)$$

(we leave the proof as an exercise).

These results suggest that (U, V) has a bivariate normal distribution. In fact the following theorem holds:

Theorem 13.4.1 *The estimators (V, U, T^2) are jointly sufficient for the parameter $\theta = (a, b, \sigma^2)$. Moreover, (U, V) have a bivariate normal distribution with means, variances, and covariance given by (13.12) through (13.14). Finally, T^2 is independent of (U, V) , and nT^2/σ^2 has a chi-square distribution with $n - 2$ degrees of freedom.*

Proof: We can use the joint moment generating function

$$m(t_1, t_2, t_3) = E(e^{t_1 U + t_2 V + t_3 T^2}).$$

Substituting the expression for U, V , and T^2 , and integrating with respect to the joint density of $(Y_1, \dots, Y_n)$, we can show (after considerable algebra) that $m(t_1, t_2, t_3) = m_1(t_1, t_2) \times m_2(t_3)$, which proves the independence of (U, V) and T^2 . The form of functions m_1 and m_2

will then show that claims about the distributions of (U, V) and T^2 are also valid. We omit the details. $\square$

We will now illustrate the applications of Theorem 13.4.1 with some examples on the construction of confidence intervals and tests for parameters of the regression model.

■ EXAMPLE 13.10 Is It Good to Be a Royal Prince?

Poland had altogether 13 kings whose fathers were also Polish kings (starting from the fifteenth century, Polish kings were elected, and the election of a late king's son, although often likely, was by no means ensured). In one pair, the son died young in battle; deaths in the remaining 12 pairs came from natural causes. The ages at death of fathers (x_i) and sons (y_i) are listed below.

Father	x_i	Son	y_i
Mieszko I	62	Bolesław the Brave	59
Bolesław the Brave	59	Mieszko II	44
Casimir I the Restorer	42	Bolesław the Bold	42
Władysław I the Short	73	Casimir III the Great	60
Władysław II Jagiełło	83	Casimir IV Jagiellonian	65
Casimir IV Jagiellonian	65	John I Albert	42
Casimir IV Jagiellonian	65	Alexander Jagiellonian	45
Casimir IV Jagiellonian	65	Sigismund the Old	81
Sigismund I the Old	81	Sigismund II Augustus	52
Sigismund III Vasa	66	Władysław IV Vasa	53
Sigismund III Vasa	66	John II Casimir Vasa	63
Augustus II the Strong	63	Augustus III	67

As can be seen, only in 2 out of 12 pairs did the son live longer than his father. Can this be attributed to chance, or does it indicate some systematic trend (e.g., a “pampered” heir to the crown is not capable of dealing with the stress and additional duties demanded of him upon his accession to the crown).

We have here

$$\bar{x} = 65.83, \quad \bar{y} = 56.08$$

which means that on average, sons lived about 10 fewer years than their fathers. For the regression coefficients, and their variances and covariances, we obtain

$$\hat{a} = 29.214, \quad \hat{b} = 0.408,$$

and also $\sum x_i^2 = 53,224$, $\sum (x_i - \bar{x})^2 = 1,215.67$. Finally, $T^2 = 115.034$. The individual points, as well as the regression line $y = 29.214 + 0.408x$, are presented in Figure 13.3 (broken line). We will return to the analysis in Example 13.11.

We begin by constructing confidence intervals for the regression intercept a and slope b . It is known that $(U - b)/\sigma_U$ has a standard normal distribution. This quantity, however, involves the nuisance parameter σ . Since nT^2/σ^2 is independent of U and has a chi-square distribution with $n - 2$ degrees of freedom, the ratio

$$t = \frac{(U - b)/\sigma_U}{\sqrt{nT^2/[\sigma^2(n - 2)]}} = \frac{(U - b)\sqrt{(n - 2)\sum (x_i - \bar{x})^2}}{\sqrt{\sum (Y_{x_i} - Ux_i - V)^2}} \quad (13.15)$$

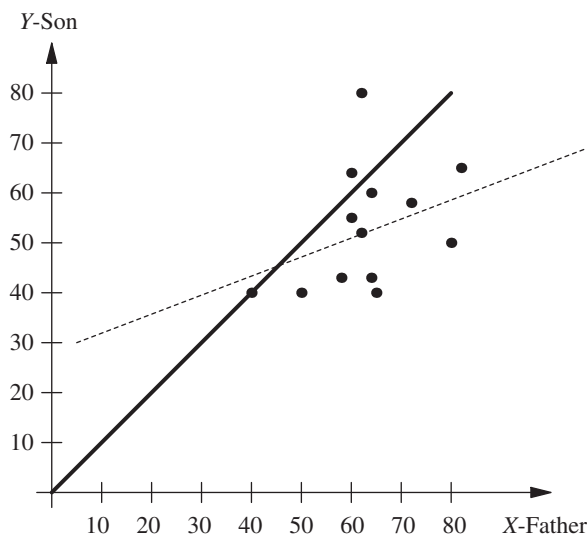


Figure 13.3 Ages of Polish kings and their heirs at death.

has a Student's t distribution with $n - 2$ degrees of freedom. Consequently, (13.15) is a pivotal quantity for b (see Definition 11.7.2). Letting $\hat{a}$ and $\hat{b}$ denote the observed values of estimators V and U of intercept and slope, we obtain the $(1 - \gamma)$ -level confidence interval for slope b as

$$\hat{b} \pm \frac{t_{\gamma/2, n-2}}{\sqrt{(n-2) \sum (x_i - \bar{x})^2}} \sqrt{\sum (y_j - \hat{a} - \hat{b}x_j)^2}.$$

In a similar way, we can derive the $(1 - \gamma)$ -level confidence interval for the intercept a in regression line:

$$\hat{a} \pm t_{\gamma/2, n-2} \sqrt{\frac{\sum x_i^2}{(n-2) \sum (x_i - \bar{x})^2}} \sqrt{\sum (y_i - \hat{a} - \hat{b}x_i)^2}.$$

■ EXAMPLE 13.11

To continue Example 13.10, the 95% confidence intervals for the regression slope and the intercept are now, respectively,

$$[0.408 - 0.751; 0.408 + 0.751] = [-0.343, 1.159]$$

and

$$[29.214 - 50.000; 29.214 + 50.000] = [-20.786; 79.214].$$

Since the confidence interval for regression slope b covers the value 0, we cannot exclude the possibility that the true regression is a constant a . This means that the age of the son at death does not depend on the father's age at death. Thus, the evidence is not conclusive with regard to the effect of being born to a royal family on duration of life.

Obviously, this example should not be taken too seriously. There are many factors affecting the length of one's life that should also be taken into account in a real study.

We can even construct a simultaneous confidence set for both regression parameters using the F distribution. We want to show (we omit the proof) that the random variable

$$Q = \frac{1}{\sigma^2} \left[n(U - b) + 2n\bar{x}(U - b)(V - a) + \sum x_i^2(V - a)^2 \right]$$

has the chi-square distribution with 2 degrees of freedom. Consequently, the random variable

$$\frac{Q/2}{nT^2/(n-2)} = \frac{n-2}{2} \times \frac{[n(U - b) + 2n\bar{x}(U - b)(V - a) + \sum x_i^2(V - a)^2]}{\sum (Y_{x_i} - Ux_i - V)^2}$$

has the F distribution with $(2, n-2)$ degrees of freedom. Thus, the ellipsoid in the (a, b) -plane,

$$n(\hat{b} - b)^2 + 2n\bar{x}(\hat{b} - b)(\hat{a} - a) + \sum x_i^2(\hat{a} - a)^2 \leq \frac{2}{n-2} F_{\gamma, 2, n-2} \sum (y_i - \hat{a} - \hat{b}x_i)^2,$$

is a $(1 - \gamma)$ -level confidence set for the regression parameters.

Notice that the results obtained thus far can be used to build estimators and to construct tests of hypotheses about linear combinations of the regression coefficients a and b . We will illustrate the situation with an example.

■ EXAMPLE 13.12

Suppose that we need to estimate the parameter $\theta = Aa + Bb$, where A and B are given constants. A special case is obtained here if $A = 1, B = x_0$, so that the objective is to estimate the mean response at $X = x_0$, namely $E(Y_{x_0}) = a + bx_0$.

Clearly, the unbiased estimator of θ is $W = AV + BU$, whose value for the sample is $\hat{\theta} = A\hat{a} + B\hat{b}$. The distribution of W is normal since W is a sum of two normally distributed random variables. So we have

$$\begin{aligned} \sigma_W^2 &= A^2\sigma_V^2 + B^2\sigma_U^2 + 2ABCov(U, V) \\ &= \frac{\sigma^2}{\sum (x_i - \bar{x})^2} \left[A^2 \sum x_i^2 - 2AB\bar{x} + B^2 \right]. \end{aligned}$$

Proceeding as before, we can show that the random variable

$$\frac{[A(V - a) + B(U - b)]\sqrt{(n-2)\sum (x_i - \bar{x})^2}}{\sqrt{[A^2 \sum x_i^2 - 2AB\bar{x} + B^2] \sum (Y_{x_i} - Ux_i - V)^2}}$$

has the Student's t distribution with $n-2$ degrees of freedom. This gives the $(1 - \gamma)$ -level confidence interval for θ as

$$\hat{\theta} \pm t_{\gamma/2, n-2} \sqrt{\frac{A^2 \sum x_i^2 - 2AB\bar{x} + B^2}{(n-2) \sum (x_i - \bar{x})^2} \sum (y_i - \hat{a} - \hat{b}x_i)^2}.$$

These results can also be used to test hypotheses about the regression coefficients a and b . The testing procedures (likelihood ratio tests) use the same Student-type ratios as the confidence intervals above. We will give the results for the slope coefficient b , leaving the derivation of the tests for intercept a as exercises.

Suppose that we want to test the null hypothesis

$$H_0 : b = b_0$$

against a one- or two-sided alternative. Then, given H_0 , the random variable $(U - b_0)/\sigma_U$ has a standard normal distribution, and consequently the test statistic obtained upon division by $\sqrt{nT^2/[\sigma^2(n-2)]}$, that is,

$$t = \frac{U - b_0}{\sqrt{\sum (Y_{x_i} - Ux_i - V)^2}} \sqrt{(n-2) \sum (x_i - \bar{x})^2}$$

has t distribution with $n - 2$ degrees of freedom. A one- or two-sided rejection region is then used depending on the alternative hypothesis.

PROBLEMS

13.4.1 Derive the test for the null hypothesis $H_0 : a = a_0$ against the one- or two-sided alternative.

13.4.2 The scores on an entrance exam (x) and the grade point average (GPA's) upon graduation (y) for 10 randomly selected students of a certain university are

x	355	361	402	365	375	404	349	380	420	395
y	3.66	3.49	3.86	3.24	3.55	3.92	3.11	3.19	3.76	3.75

Assume normality and homoscedasticity. **(i)** Compute the MLE's of a , b , and σ^2 . **(ii)** Test the hypothesis that there is no relation between the grade on the entrance exam and the GPA, against the alternative that higher scores on the entrance exam tend to be associated with higher GPA's. **(iii)** Find the shortest 95% confidence intervals for a and for b . **(iv)** Find the joint confidence set for (a, b) with confidence level 0.95; sketch it and compare with your answers to (iii).

13.4.3 Suppose that observations are taken only at two values, x_1 and x_2 , of an independent variable. Let $\bar{y}_1$ and $\bar{y}_2$ be the average observed responses for $x = x_1$ and for $x = x_2$, respectively. Show that the estimated regression line passes through points $(x_1, \bar{y}_1)$ and $(x_2, \bar{y}_2)$.

13.4.4 Using the ideas given on deriving the confidence set for (a, b) , derive the testing procedure for the null hypothesis $H_0 : a = a_0, b = b_0$ against the general alternative $H_1 : H_0$ is false. Consider two cases: **(i)** σ known. **(ii)** σ unknown.

13.4.5 Suppose it is known that the true regression (assuming a normal case) has the form $E(Y_x) = bx$. Derive the MLE's for b and for σ^2 .

13.5 TESTING LINEARITY

In this section, we make the following two additional assumptions regarding the design of the experiment for collecting the data on regression parameters:

- (i) There are at least three distinct values among $x_1, x_2, \dots, x_n$.
- (ii) There exist at least two repeated values among $x_1, \dots, x_n$.

We will show that under (i) and (ii) it is possible to construct a test for linearity of regression, that is, a test of the null hypothesis

$$H_0 : E(Y_x) = a + bx \text{ for some } a, b \quad \text{against} \quad H_1 : E(Y_x) = u(x),$$

where $u(x)$ is not a linear function of x .

The test for linearity will be based on construction of two independent unbiased estimators of σ^2 . One of them will estimate σ^2 regardless of whether or not H_0 is true, and the other will be an unbiased estimator of σ^2 only under the null hypothesis.

For the considerations of the present section only, it will be convenient to change the labeling of the sample (x_i, y_i) , $i = 1, \dots, n$, as follows: Let $x'_1, x'_2, \dots, x'_r$ be all those x_i at which multiple observations were made, and let $n_1, \dots, n_r$ be the numbers of observations made for those values. Furthermore, let the observations made for x'_j be $y'_{j,1}, \dots, y'_{j,n_j}$, and regarded as values of iid random variables $Y'_{j,1}, \dots, Y'_{j,n_j}$. Finally, the remaining values of independent variable will be denoted by $x'_{r+1}, \dots, x'_m$, with the corresponding observations $y'_{r+1}, \dots, y'_m$ being the values of random variables $y'_{r+1}, \dots, y'_m$.

We have therefore

$$n_1 + \dots + n_r + (m - r) = n, \quad (13.16)$$

with $n_i \geq 2, i = 1, \dots, r$.

Consider now the following decomposition of the sum of the squared deviations:

$$\begin{aligned} s^2 &= \sum_{i=1}^n (y_i - \hat{a} - \hat{b}x_i)^2 = \sum_{j=1}^r \sum_{t=1}^{n_j} (y'_{j,t} - \hat{a} - \hat{b}x'_j)^2 + \sum_{k=r+1}^m (y'_k - \hat{a} - \hat{b}x'_k)^2 \\ &= \sum_{j=1}^r \sum_{t=1}^{n_j} (y'_{j,t} - \bar{y}'_j + \bar{y}'_j - \hat{a} - \hat{b}x'_j)^2 + \sum_{k=r+1}^m (y'_k - \hat{a} - \hat{b}x'_k)^2 \\ &= \sum_{j=1}^r \sum_{t=1}^{n_j} (y'_{j,t} - \bar{y}'_j)^2 + \sum_{j=1}^r n_j (\bar{y}'_j - \hat{a} - \hat{b}x'_j)^2 + \sum_{k=r+1}^m (y'_k - \hat{a} - \hat{b}x'_k)^2 \\ &= s_1^2 + s_2^2 + s_3^2. \end{aligned}$$

In passing from the third to the fourth expression, the cross-products were omitted. One can check that indeed they are equal to zero.

Now the sums s_1^2 , s_2^2 , and s_3^2 are observed values of random variables

$$\begin{aligned} S_1^2 &= \sum_{j=1}^r \sum_{t=1}^{n_j} (Y'_{j,t} - \bar{Y}'_j)^2, \\ S_2^2 &= \sum_{j=1}^r n_j (\bar{Y}'_j - V - Ux'_j)^2, \\ S_3^2 &= \sum_{k=r+1}^m (Y'_k - V - Ux'_k)^2. \end{aligned}$$

Under the assumption of normality of the distributions of Y_x , the random variables S_1^2 and S_2^2 are independent; random variables s_3^2 and S_1^2 are independent as well. The proof of these facts is similar to the proof of Theorem 9.2.1, and it will be omitted.

Finally, in view of (13.16), S_1^2/σ^2 has a chi-square distribution with the number of degrees of freedom equal to

$$\sum_{j=1}^r (n_j - 1) = \sum_{j=1}^r n_j - r = n - m,$$

where $n - m > 0$, since $r \geq 1$ and $n_j \geq 2$ for all j . It is important to recognize that this statement about the distribution of S_1^2 holds *regardless of whether the null hypothesis H_0 about linearity of regression is true.*

On the other hand, if H_0 is true, then also

$$S_2^2/\sigma^2 + S_3^2/\sigma^2 \sim \chi_{m-2}^2.$$

Again, the number of degrees of freedom is positive, in view of the assumption that m , the number of distinct values of x_i , is at least 3.

To obtain the testing procedure, observe finally that any violations of the null hypothesis will tend to increase the expected value of the sum $S_2^2 + S_3^2$, since $E(Y - \xi)^2$ is minimized for $\xi = E(Y)$.

Consequently,

$$F = \frac{(S_2^2 + S_3^2)/(m-2)}{S_1^2/(n-m)}$$

has the F -distribution with $(m-2, n-m)$ degrees of freedom, provided that H_0 is true, while any lack of fit to the linear model will tend to inflate the value of F . Thus, the testing procedure with size α is as follows: reject the hypothesis of linearity of regression if

$$\frac{\left[\sum_{j=1}^r n_j (\bar{y}'_j - \hat{a} - \hat{b}x'_j)^2 + \sum_{k=r+1}^m (y'_k - \hat{a} - \hat{b}x'_k)^2 \right] / (m-2)}{\sum_{j=1}^r \sum_{t=1}^{n_j} (y'_{j,t} - \bar{y}'_j)^2 / (n-m)} > F_{\alpha, m-2, n-m}.$$

■ EXAMPLE 13.13

To develop some sort of intuition concerning the linearity test, we will analyze the situation in a deliberately oversimplified case: two observations for $x = 0$ are d and $-d$, an observation for $x = 1$ is c , and an observation for $x = 2$ is 0. We will find the range of values c (for fixed d), and the range of values d (for given c) when the linearity hypothesis should be rejected, on a level of significance $\alpha = 0.05$, say.

Intuitively, for a fixed d linearity will be rejected if the middle point deviates too far from the x -axis in any direction, and (for fixed $c \neq 0$) when d is close to 0.

We have here $\bar{x} = (0 + 0 + 1 + 2)/4 = 3/4$ and $\sum (x_i - \bar{x})^2 = 11/4$. Moreover, $\bar{y} = (d - d + c + 0)/4 = c/4$. This gives

$$\hat{b} = \frac{c}{11}, \quad \hat{a} = \frac{2c}{11}.$$

The estimated regression line is therefore $\hat{a} + \hat{b}x = (c/11)(2 + x)$. To compute the F ratio, we have $s_1^2 = 2d^2$ with $n - m = 1$. For the numerator we find

$$s_2^2 + s_3^2 = 2 \left[0 - \frac{c(2+0)}{11} \right]^2 + \left[c - \frac{c(2+1)}{11} \right]^2 + \left[0 - \frac{c(2+2)}{11} \right]^2 = \frac{84}{121} c^2,$$

with $m - 2 = 1$ degree of freedom. Since $F_{\alpha, 1, 1} = 161.45$, the hypothesis of linearity should be rejected if

$$\frac{\frac{84}{121} c^2}{2d^2} > 161.45 \quad \text{or} \quad \frac{c^2}{d^2} > 443.99.$$

Since $d > 0$, this is equivalent to the inequality $|c| > 21.56d$.

PROBLEMS

- 13.5.1** The output of a certain device is suspected to decrease linearly with the temperature. Two observations were taken for each temperature, and the data (in appropriate units) are as follows:

Temperature	55	65	75	85	95	105
Observation 1	2.01	2.01	2.02	1.48	1.93	1.90
Observation 2	2.03	2.02	2.00	1.48	1.95	1.94

At the significance level 0.05, test the hypothesis that the output is a linear function of temperature.

13.5.2 Suppose that we have six data points in addition to those in Problem 13.4.2:

x	355	402	402	309	375	375
y	3.44	3.91	3.95	3.24	3.52	3.31

Test the hypothesis (using all 17 data values) that the regression of GPA's on the scores from the entrance exam is linear.

13.5.3 Check the identity (13.17).

13.5.4 Derive, if possible, a test of linearity of regression under the assumptions of this section, if the individual observations for values $x'_{r+1}, \dots, x'_m$ are now unknown, and instead we have the data on averages $y'_{r+1}, \dots, y'_m$, and the corresponding numbers of observations $n'_{r+1}, \dots, n'_m$.

13.6 PREDICTION

Consider now the problem of prediction. As before, the data have the form of a set of pairs $(x_i, y_i), i = 1, \dots, n$, where y_i is the observed value of a random variable Y_{x_i} assumed to be normal with mean $a + bx_i$ and standard deviation σ . The random variables $Y_{x_1}, \dots, Y_{x_n}$ are independent, and at least two among $x_1, \dots, x_n$ are distinct. The problem is to predict Y_{x_0} as precisely as possible.

More generally, we want to predict the average of k independent observations of Y_{x_0} . By prediction, we mean here providing an interval, as short as possible, such that the value of the predicted random variable will fall into this interval with a preassigned probability, say $1 - \gamma$. Let $\bar{Y}_{x_0}$ denote the average of k observations to be taken at the value x_0 of the independent variable. We are looking for an interval $[c_1, c_2]$ such that

$$P\{c_1 \leq \bar{Y}_{x_0} \leq c_2\} = 1 - \gamma.$$

The solution will be obtained as follows. Note first that the average $\bar{Y}_{x_0}$ has distribution $N(a + bx_0, \sigma^2/k)$. Consequently,

$$Z = \frac{\bar{Y}_{x_0} - a - bx_0}{\sigma/\sqrt{k}}$$

is a standard normal random variable, and we have

$$\{-z_{\gamma/2} \leq Z \leq z_{\gamma/2}\} = 1 - \gamma.$$

A simple argument based on symmetry around 0 of normal density shows that $(-z_{\gamma/2}, z_{\gamma/2})$ is the shortest prediction interval for Z . Thus, the corresponding shortest prediction interval for $\bar{Y}_{x_0}$ (given that the regression parameters a, b , and σ^2 are known) is

$$a + bx_0 \pm z_{\gamma/2} \frac{\sigma}{\sqrt{k}},$$

and its length is $2(\sigma/\sqrt{k})z_{\gamma/2}$. The actual prediction interval has to take into account the fact that the regression parameters are estimated. The construction is based on an analogue to the pivotal quantity. Thus, the random variable

$$L = \bar{Y}_{x_0} - Ux_0 - V$$

has a normal distribution (being a linear combination of normal random variables), and $E(L) = 0$ in view of the fact that U and V are unbiased estimators of b and a . We have, using the fact that $\bar{Y}_{x_0}$ is independent of (U, V) ,

$$\begin{aligned}\sigma_L^2 &= \text{Var}(L) = \text{Var}(\bar{Y}_{x_0}) + x_0^2 \text{Var}(U) + \text{Var}(V) + 2x_0 \text{Cov}(U, V) \\ &= \frac{\sigma^2}{k} + x_0^2 \sigma_U^2 + \sigma_V^2 + 2x_0 \text{Cov}(U, V) \\ &= \frac{\sigma^2}{k} + x_0^2 \frac{\sigma^2}{\sum (x_i - \bar{x})^2} + \frac{\sigma^2 \sum x_i^2}{n \sum (x_i - \bar{x})^2} - 2x_0 \frac{\sigma^2 \bar{x}}{\sum (x_i - \bar{x})^2} \\ &= \sigma^2 \left[\frac{1}{k} + \frac{x_0^2 + \frac{1}{n} \sum x_i^2 - 2x_0 \bar{x}}{\sum (x_i - \bar{x})^2} \right] = \sigma^2 \left[\frac{1}{k} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right].\end{aligned}$$

Consequently, the random variable

$$\frac{L}{\sigma_L} = \frac{\bar{Y}_{x_0} - Ux_0 - V}{\sigma \sqrt{1/k + 1/n + (x_0 - \bar{x})^2 / \sum (x_i - \bar{x})^2}}$$

has a standard normal distribution. Dividing by $\sqrt{nT^2/[\sigma^2(n-2)]}$, we obtain

$$t = \frac{(\bar{Y}_{x_0} - Ux_0 - V)\sqrt{n-2}}{\sqrt{1/k + 1/n + (x_0 - \bar{x})^2 / \sum (x_i - \bar{x})^2} \sqrt{\sum (\bar{Y}_{x_0} - Ux_i - V)^2}},$$

which has the t -distribution with $n-2$ degrees of freedom. Substituting the observed values $\hat{a}$ and $\hat{b}$ of V and U , we obtain the prediction interval for $\bar{Y}_{x_0}$ with prediction probability $1 - \gamma$ as

$$\hat{a} + \hat{b}x_0 \pm \frac{t_{\gamma/2, n-2}}{\sqrt{n-2}} \sqrt{\left[\frac{1}{k} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] \left[\sum (y_{x_i} - \hat{a} - \hat{b}x_i)^2 \right]}. \quad (13.17)$$

Let us remark here that the prediction interval (for fixed n and k) is shortest if $x_0 = \bar{x}$ (i.e., it is “easier” to predict values of dependent variable for x_0 close to $\bar{x}$).

Observe also that as $k \rightarrow \infty$, the length for prediction interval for *known* a, b , and σ tends to 0. In the present case, the increase of k has much less effect, and as $k \rightarrow \infty$, the length of the prediction interval tends to a positive quantity depending on n and on the location of observations $x_1, x_2, \dots, x_n$. This is consistent with intuition, according to which in the present case the uncertainty of prediction has two sources: randomness of the dependent variable about its estimated mean and uncertainty as to the exact location of the true mean.

PROBLEMS

13.6.1 At harvest, the weight of a certain fruit grown in a greenhouse has the $N(a + bt, \sigma^2)$ distribution, where t is the average temperature. Weights in a sample of five fruits for $t = 80^\circ\text{F}$ are 1.02, 1.03, 0.98, 1.05, 1.02, while a sample of seven fruits for $t = 86^\circ\text{F}$ (other conditions being equal) are 1.03, 1.03, 1.09, 1.07, 1.04, 1.02, 1.08. Give a 95% prediction interval for the average weight of four fruits grown: **(i)** In the first greenhouse. **(ii)** In the second greenhouse. **(iii)** In a greenhouse for $t = 84^\circ\text{F}$.

- 13.6.2** Combine data points in Problems 13.4.2 and 13.5.2, and find the 95% prediction interval for the GPA of a student who scored 400 on an entrance exam.
- 13.6.3** Find the prediction interval with a probability $1 - \gamma$ of coverage for an observation to be taken at the value x_0 by an independent variable, given the data (x_i, y_i) , $i = 1, \dots, n$, and assuming the normal model of the form $Y_x = bx + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$.
- 13.6.4** Under condition of Problem 13.6.3, find the prediction interval for the mean of k observations taken for value x_0 of the independent variable.

13.7 INVERSE REGRESSION

Inverse regression is the problem of inference about an unknown value x_0 of an independent variable on the basis of a number of observations of a response for this value. The data consist of two groups of observations. One group is, as before, the sample

$$(x_i, y_i), \quad i = 1, \dots, n,$$

where y_i is the observed value of random variable $Y_i \sim N(a + bx_i, \sigma^2)$, with the usual assumption of independence. The second group, $(y'_1, \dots, y'_m)$, is a random sample from an $N(a + bx_0, \sigma^2)$ distribution, where x_0 is unknown. The objective is to estimate x_0 , with a, b , and σ^2 being unknown.

The likelihood of the data is

$$\begin{aligned} L &= \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\sigma^2(y_i - a - bx_i)^2} \prod_{j=1}^m \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\sigma^2(y'_j - a - bx_0)^2} \\ &= C(\sigma^2)^{-(m+n)/2} \exp \left\{ -\frac{\sum_{i=1}^n (y_i - a - bx_i)^2 + \sum_{j=1}^m (y'_j - a - bx_0)^2}{2\sigma^2} \right\}. \end{aligned}$$

Taking the logarithm and differentiating with respect to a, b, σ^2 , and x_0 , we obtain the equations

$$\begin{aligned} \sum_{i=1}^n (y_i - a - bx_i) + \sum_{j=1}^m (y'_j - a - bx_0) &= 0, \\ \sum_{i=1}^n (y_i - a - bx_i)x_i + x_0 \sum_{j=1}^m (y'_j - a - bx_0) &= 0, \\ \sum_{i=1}^n (y_i - a - bx_i)^2 + \sum_{j=1}^m (y'_j - a - bx_0)^2 - (m+n)\sigma^2 &= 0, \\ \sum_{j=1}^m (y'_j - a - bx_0) &= 0. \end{aligned}$$

Letting $\bar{y}' = (1/m) \sum_{j=1}^m y'_j$, we obtain from the last equation the estimate

$$\hat{x}_0 = \frac{\bar{y}' - \hat{a}}{\hat{b}}, \quad (13.18)$$

assuming, of course, that $\hat{b} \neq 0$. The third equation gives

$$\hat{\sigma}^2 = \frac{1}{m+n} \left[\sum_{i=1}^n (y_i - \hat{a} - \hat{b}x_i)^2 + \sum_{j=1}^m (y_j - \bar{y}')^2 \right].$$

Using (13.18) in the first two equations, we can easily check that in each case the sum involving y'_j equals 0, which means that the expressions for $\hat{a}$ and $\hat{b}$ are given by (13.9).

These results are consistent with our intuition. Indeed, since we do not know x_0 , the observations $y'_1, \dots, y'_m$ cannot provide any information about the slope and intercept of the regression line. On the other hand, if only $m > 1$, the values $y'_1, \dots, y'_m$ provide additional information about σ^2 .

To set a confidence interval for $\hat{x}_0$, we may proceed as in the case of a prediction. The random variable $W = \bar{Y}' - Ux_0 - V$ has the normal distribution with mean 0 and variance

$$\sigma_W^2 = \sigma^2 \left(\frac{1}{m} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_0 - \bar{x})^2} \right).$$

Consequently, W/σ_W has a standard normal distribution. To eliminate σ , we note that the random variable

$$\frac{(m+n)\hat{\sigma}^2}{\sigma^2} = \frac{1}{\sigma^2} \left[\sum_{i=1}^n (Y_{x_i} - Ux_i - V)^2 + \sum_{j=1}^m (Y'_{x_{0,j}} - \bar{Y}')^2 \right]$$

has the chi-square distribution with $m+n-3$ degrees of freedom. Indeed, the two sums are independent, with $n-2$ and $m-1$ degrees of freedom, respectively. Consequently, the random variable

$$\frac{W/\sigma_W}{\sqrt{(m+n)\hat{\sigma}^2/\sigma^2(m+n-3)}} \quad (13.19)$$

has the Student's t distribution with $m+n-3$ degrees of freedom. The observed value of the random variable given by (13.19) is

$$t = \frac{(\bar{y}' - \hat{a} - \hat{b}x_0)\sqrt{m+n-3}}{\sqrt{\frac{1}{m} + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_0 - \bar{x})^2} \times \sum_{i=1}^n (y_i - \hat{a} - \hat{b}x_0)^2 + \sum_{j=1}^m (y'_j - \bar{y}')^2}},$$

and the confidence interval is obtained by converting the inequality

$$-t_{\gamma/2, m+n-3} < t < t_{\gamma/2, m+n-3} \quad (13.20)$$

into an inequality for x_0 . Observe, however, that (13.20) is now a *quadratic* inequality.

■ EXAMPLE 13.14

The amounts of a chemical compound (y) that dissolve in a given amount of water at different temperatures (x) are given in the following table:

x (°C)	y (grams)		
5	3	4	2
10	7	7	6
15	10	13	11
20	15	18	17
25	21	18	19

Two measurements for an unknown temperature x_0 are 14 and 16. What can one say about x_0 ?

SOLUTION. We have here $n = 15$ and $m = 2$. The relevant quantities are

$$\hat{a} = -1.14, \quad \hat{b} = 0.853, \quad \bar{x} = 15, \quad \sum (x_i - \bar{x})^2 = 750,$$

$$\sum (y_i - \hat{a} - bx_i)^2 = 21.46 \quad \bar{y}' = 15, \quad \sum (y'_i - \bar{y}')^2 = 2.$$

Thus, the point estimate of x_0 is $\hat{x}_0 = 19.23$. Since $t_{0.05,14} = 2.145$ ($n + m - 3 = 14$), to obtain a 95% confidence interval for x_0 , we must solve the inequality (13.20), which in our case takes the form

$$\frac{|15 - 0.853x_0 + 1.4|}{\sqrt{\frac{1}{2} + \frac{1}{15} + (x_0 - 15)^2/750}} \frac{\sqrt{2 + 15 - 3}}{\sqrt{21.46 + 2}} < 2.145$$

or

$$\frac{|346.98 - 18.05x_0|}{\sqrt{x_0^2 - 30x_0 + 650}} < 2.145.$$

After some algebra, we obtain $x_0^2 - 38.567x_0 + 365.499 < 0$, which gives the confidence interval

$$16.96 < x_0 < 21.80.$$

PROBLEMS

13.7.1 Five measurements of Y at $x = 10$ are 10.5, 10.6, 9.7, 11.1, and 12.3. Six measurements of Y at $x = 20$ are 3.1, 3.6, 3.1, 4.0, 5.2, and 2.9. Assuming that the regression of Y on X is linear, estimate the value of x if two observations made at this value are 6.3 and 7.1.

13.7.2 Using data from Problems 13.4.2 and 13.5.2, estimate the score on an entrance exam of a student who graduated with a GPA equal to 3.95.

13.8 BLUE

Most of the results presented thus far rely on the assumptions that the random variable Y_x is normally distributed with constant variance and that the observations are independent. These assumptions allow us to use the likelihood and provide access to the distributions of MLE's. The natural question is what to do if the assumptions above are not satisfied.

First, with regard to homoscedasticity, there exist numerous variance-stabilizing transformations. These transformations have been suggested by statisticians as an ad hoc remedy

against heteroscedasticity: instead of data of the form (x_i, y_i) , one can use the data (x_i, y_i^*) , with $y_i^* = g(y_i)$, where g is some suitably selected function.

One can also use transformation of y 's that depend on x 's, that is, by replacing the pair (x_i, y_i) with $(x_i, g_{x_i}(y_i))$, where g_x is some function [e.g., replacing (x_i, y_i) by $(x_i, y_i/x_i)$]. Which transformation should be used in a given situation can be hard to resolve, especially when little is known about the distribution of Y_x . It is rather statistical intuition and experience that can serve as a guide.

A question interesting from a both theoretical and practical point of view is what to do if the distribution of Y_x is not normal. Suppose, for instance, that the model analyzed is that of linear regression $Y_x = a + bx + \epsilon_x$, where the errors ϵ_x can be assumed to satisfy the conditions;

- (i) $E(\epsilon_x) = 0$ (unbiasedness).
- (ii) $E(\epsilon_x^2) = \sigma^2$ (homoscedasticity).
- (iii) $E(\epsilon_x \epsilon_{x'}) = 0$ (orthogonality).

The least squares estimators of regression coefficients a and b (which are also the MLE's in the normal case) are

$$\hat{b} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x})y_i}{\sum (x_i - \bar{x})^2} \quad \text{and} \quad \hat{a} = \bar{y} - \hat{b}\bar{x}.$$

The estimator of σ^2 may be based on residuals, where

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{j=1}^n (y_j - \hat{a} - \hat{b}x_j)^2$$

is the observed value of the estimator

$$S^2 = \frac{nT^2}{(n-2)} = \frac{1}{n-2} \sum_{j=1}^n (Y_j - V - U_{x_j})^2,$$

with U , V , and T given by (13.11). Clearly, U , V , and S^2 are unbiased estimators of b , a , and σ^2 , with variances and covariances obtained as before. This is true because the calculations of moments did not rely on the assumption of normality.

It is possible to show that under the assumption of normality, U and V are also minimum variance estimators of b and a (i.e., that their variances coincide with the bounds given by the Rao-Cramèr inequality).

The question is: Is there a reason to use U and V as estimators of b and a when the normality assumption does not hold? The mere availability of the computational formulas in statistical packages is hardly a justification. Unbiasedness of U and V is a desirable property only if it can be related to MSE. As we have seen, there are situations where the use of biased estimators is recommended, as leading to smaller mean square error.

The answer is positive and is given by the following theorem (which we state here without a proof).

Theorem 13.8.1 (Gauss–Markov) *Consider the observations (x_i, Y_j) , $i = 1, 2, \dots, n$, with $Y_i = a + bx_i + \epsilon_i$, where the errors ϵ_i satisfy conditions (i)–(iii) of unbiasedness, homoscedasticity, and orthogonality. Let $\mathcal{L}$ be the class of all statistics of the form*

$$r_1 Y_1 + r_2 Y_2 + \dots + r_n Y_n,$$

where r_i are constants—depending possibly on the vector $(x_1, \dots, x_n)$. Furthermore, let $\mathcal{L}_a \subset \mathcal{L}$, $\mathcal{L}_b \subset \mathcal{L}$ be the subsets of $\mathcal{L}$ consisting of statistics that are unbiased estimators of a and of b . Then, the statistics V and U given by (13.11) have minimal variances in classes $\mathcal{L}_a$ and $\mathcal{L}_b$.

The acronym used here is BLUE, which stands for the “best linear unbiased estimator.” Thus, V is BLUE for a , and U is BLUE for b .

Note that in view of unbiasedness, “best” estimators mean those with a minimum mean square error. Recall that in normal case U and V are best (in the sense of MSE) estimators of regression parameters in the class of *all* estimators, linear or not. In the present case, under the weaker assumption the conclusion is also weaker: U and V are best estimators in a more restricted class of estimators, namely those that depend linearly on the data.

PROBLEMS

13.8.1 Suppose that the number of errors in a text of length x is known to be a Poisson random variable with unknown mean λ . We observe n texts of lengths $x_1, x_2, \dots, x_n$ and find the numbers of errors they contain, $Y_1, \dots, Y_n$, satisfy $E(Y_j) = \text{Var}(Y_j) = \lambda x_j$. Find the BLUE of λ .

13.8.2 Carry out the calculations in the following direct proof of the Gauss–Markov theorem showing that the LS estimators of a and b are BLUE. For a you need to determine the constants $a_i, i = 1, \dots, n$ such that the statistic $T = \sum_{i=1}^n a_i Y_i$ satisfies the conditions: (1) $E(T) = a$. (2) Variance of T is the smallest among all linear estimators for which condition (1) holds. Show first that condition (1) implies

$$\sum a_j = 1, \quad \sum a_j x_j = 0. \quad (13.21)$$

Next use the fact that Y_j 's are uncorrelated and homoscedastic, and show that $\text{Var}(T) = \sigma^2 \sum a_j^2$, that is, minimize $\sum a_j^2$ subject to constraints (13.21). Using Lagrange multipliers, this means that one must minimize

$$\sum a_j^2 - \lambda_1 \sum (a_j - 1) - \lambda_2 \sum a_j x_j.$$

Take the derivatives with respect to $a_1, \dots, a_n, \lambda_1$, and λ_2 , solve the resulting $n + 2$ equations, and check that the solution a_j gives the LS estimator of a .

13.8.3 Provide the same argument as in Problem 13.8.2 for the LS estimator of b .

13.8.4 Given data $(x_i, y_i), i = 1, \dots, n$, find the LS estimators of the quadratic regression $a + bx + cx^2$.

13.9 REGRESSION TOWARD THE MEAN

The phenomenon known as *regression toward the mean* was discovered in the nineteenth century by Galton, who studied various hereditary traits. He noticed that (e.g., in using height) tall fathers tend to have tall sons, but their sons tend to be closer to the average than the fathers. Similarly, short fathers tend to have short sons, but their sons tend to be closer to the average than their fathers. Galton called it a “tendency toward mediocrity.” Galton’s choice of the word “mediocrity” may also explain why he chose the term “regression,” a word with somewhat derogatory connotations.

It should be realized that regression, understood as “affecting the mean of dependent variable Y by independent variable X ,” need not imply any casual relationship between the values of X and Y . One of the more common types of relations between X and Y , that leads to the regression phenomenon and yet does not involve any casual effects of X on Y , is exemplified by the following situation: Imagine some attribute of objects, for example, length. Suppose that a person takes two observations, typically differing somewhat

because of a measurement error, and calls the first and second measurement of the i th object x_i and y_i .

Imagine now that points (x_i, y_i) are plotted, resulting in a scatter diagram. If the measurement errors are small, and/or the objects measured differ in their true lengths, we will observe that the points (x_i, y_i) have a strong linear relationship with a slope close to 1.

Such a relationship will appear stronger when the variability of lengths of measured objects is higher. This effect is utilized in designing some psychological questionnaires.

■ EXAMPLE 13.15 Psychological Test Scores

In the areas of psychology dealing with personality or motivation (as opposed to areas such as memory studies, with more quantifiable experiments), a researcher typically introduces some *construct* (“neuroticism,” “self-esteem,” etc.). Those constructs are then used to explain and/or predict some behavior. The explanation has the form of specific hypotheses, such as “persons with low self-esteem are more likely to be aggressive,” etc. In addition to a theoretical justification of such hypotheses, there arises a problem of testing them empirically. Clearly, one needs here a tool for measuring the level of the construct (a tool to measure the level of neuroticism, self-esteem, etc.).

A typical tool has the form of a questionnaire and a scoring rule. For some questions, it is the answer “yes” that contributes to the total score; for some other questions it is the answer “no.” This is done in order to eliminate any bias arising from a possible tendency toward some type of answers.

When the questionnaire is applied to a subject s , one obtains the score X_s . Upon repetition, the score may be different, say X'_s . One of the central assumptions of the theory of psychological tests is that the expected scores X_s and X'_s are equal:

$$E(X_s) = E(X'_s) = T_s.$$

Moreover, the deviations $\epsilon_s = X_s - T_s$ and $\epsilon'_s = X'_s - T_s$ are expected to satisfy the conditions

$$\text{Var}(\epsilon_s) = \text{Var}(\epsilon'_s) = \sigma_s^2, \quad E(\epsilon_s \epsilon'_s) = 0.$$

For fixed s , the value T_s (called the *true value* of the measured construct for person s) is a constant, while ϵ_s is a random variable (reflecting intra-person variability of response to the questionnaire, upon hypothetical repetitions of measurement).

Assume now that person s is sampled from some population according to a certain probability distribution. Using Theorem 7.6.4, we have

$$\begin{aligned} \sigma_X^2 &= \text{Var}(X) = E_s\{\text{Var}(X|T)\} + \text{Var}\{E_s(X|T)\} \\ &= E(\sigma_s^2) + \text{Var}(T_s) = \sigma_\epsilon^2 + \sigma_T^2. \end{aligned}$$

Clearly, $\sigma_{X'}^2 = \sigma_\epsilon^2 + \sigma_T^2$. Similarly,

$$\begin{aligned} \text{Cov}(X, X') &= E(XX') - E(X)E(X') = E\{(T + \epsilon)(T + \epsilon')\} - [E(T)]^2 \\ &= E(T^2) - [E(T)]^2 = \sigma_T^2, \end{aligned}$$

since $E(\epsilon T) = E_s\{E(\epsilon T)\} = E_s\{T_s E(\epsilon)\} = E(0) = 0$, and similarly for the other products. Thus,

$$\rho(X, X') = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}. \quad (13.22)$$

The last ratio is called the *reliability* of the test, and formula (13.22) shows that reliability is equal to the test–retest correlation. This correlation approaches 1 with an increase of σ_T^2 , that is, with an increase of variance of the true scores of the test in the population under study.

The reliability of a psychological test is (unlike of the instruments for physical measurements) not intrinsic for the test only but depends also on how diverse the population is to which the test is applied.

13.10 ANALYSIS OF VARIANCE

In the remainder of this chapter, we deal with testing for the existence of the effects of an independent variable X (often measured on the nominal scale) on a response Y .

The standard terminology of the ANOVA is that of *levels of factors*. These factors cross each other in the sense that every level of one factor can be combined with every level of the other factor.

The central assumptions of ANOVA models are very much the same as in regression models:

- (i) The response variable Y has, for each level of factors, a normal distribution with the same (unknown) variance σ^2 (*homoscedasticity*).
- (ii) The factors may affect only the mean of the response.
- (iii) Distinct observations for the same or different levels of factors are independent.

We show the tests for the hypothesis that a given factor has no effect on the response variable against the alternative that it has some effect.

The main issue here is that these tests can be carried out for various factors *on the same data*. In fact ANOVA models originated from questions arising in agriculture, where one is interested in the response variable (e.g., size of harvest Y) as dependent on combinations of various factors (e.g., type of soil, time of planting, time of harvesting, type of cultivation, and use of various fertilizers). Since a typical experiment lasts for one season, it is imperative to find a design that will allow us to study the effects of various factors using the same data. We begin with the simplest case of one factor only.

13.11 ONE-WAY LAYOUT

Consider the situation where the data are partitioned into groups, each corresponding to one level of a factor. Alternatively, the same setup may be described as “independent samples from different populations.”

We let n_i denote the number of observations from the i th group, where $i = 1, \dots, I$. We have here $I \geq 2$ and $n_i \geq 2$ for at least one i .

Let $n_1 + n_2 + \dots + n_I = N$ be the total number of observations, and let $y_{i1}, y_{i2}, \dots, y_{in_i}$ be the observations corresponding to the i th level of the factor. These observations are regarded as the recorded values of the random variables $Y_{i1}, Y_{i2}, \dots, Y_{in_i}$.

According to the assumptions stated at the beginning of this section, all random variables Y_{ij} are independent, normally distributed, with $\text{Var}(Y_{ij}) = \sigma^2$. Moreover, since the effect of a factor is expressed only through the mean, we must have

$$E(Y_{ij}) = \mu_i, \quad j = 1, \dots, n_i.$$

The objective is to test

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_I \quad \text{against} \quad H_1 : \mu_i \neq \mu_{i'}, \text{ for some } i, i'.$$

It will be convenient to let

$$\mu_i = \mu + \alpha_i, \quad i = 1, \dots, I,$$

where

$$\mu_i = \frac{\mu_1 + \dots + \mu_I}{I}. \quad (13.23)$$

We have therefore

$$\sum_{i=1}^I \alpha_i = 0,$$

and the hypotheses tested can be formulated as

$$H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_I = 0 \quad \text{against} \quad H_1 : \alpha_i \neq 0 \text{ for at least one } i.$$

Before we develop the testing procedure, let us observe that if $I = 2$, we have the problem of comparing the means of two normal populations with the same (unknown) variance σ^2 . This problem was solved in Chapter 12 (Section 12.9). The testing procedure used the t distribution with $n_1 + n_2 - 2$ degrees of freedom.

It would seem that if $I > 2$, we can use this result for the present case by comparing pairs of levels of the factor until either we find a pair where the difference is significant (and then we reject H_0) or we find no significant difference in all pairs tested (in which case we accept H_0). The reason why such a procedure is unacceptable lies in the fact that it is impossible to determine its significance level because (i) the procedures for overlapping pairs of factor levels are not independent and (ii) even for nonoverlapping pairs, if the null hypothesis is true, the chances of at least one incorrect rejection of null hypothesis increases quickly with the number of tested pairs.

Consequently, the objective is to find a procedure that can test the null hypothesis of no effect of the factor with a preassigned level of significance. The construction here will be based on a partition of the sum of squared deviations from the mean, which is very much similar to the technique used for testing linearity of regression in Section 13.6.

In the derivation below, the subscript $+$ will stand for averaging over the values of the index replaced by $+$. Thus

$$\bar{Y}_{i+} = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}, \quad \bar{y}_{i+} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij},$$

and

$$\bar{Y}_{++} = \frac{1}{N} \sum_{i=1}^I \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{N} \sum_{i=1}^I n_i \bar{Y}_{i+},$$

and similarly for $\bar{y}_{++}$.

In the identities below, we omit the cross-products. We encourage the reader to verify that all the cross-products are indeed zero. Using the fact that by (13.23) we have $E(Y_{ij}) = \mu + \alpha_i$, we decompose the sum of squared deviations of the variables from their means as follows:

$$\begin{aligned} S^2 &= \sum_{i=1}^I \sum_{j=1}^{n_i} (Y_{ij} - \mu - \alpha_i)^2 = \sum_{i=1}^I \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i+} + \bar{Y}_{i+} - \mu - \alpha_i)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i+})^2 + \sum_{i=1}^I (\bar{Y}_{i+} - \mu - \alpha_i)^2 \end{aligned} \quad (13.24)$$

$$\begin{aligned}
&= \sum_{i=1}^I \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i+})^2 + \sum_{i=1}^I n_i (\bar{Y}_{i+} - \bar{Y}_{++} + \bar{Y}_{++} - \mu - \alpha_i)^2 \\
&= \sum_{i=1}^I \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i+})^2 + \sum_{i=1}^I n_i (\bar{Y}_{i+} - \bar{Y}_{++} - \alpha_i)^2 + N(\bar{Y}_{++} - \mu)^2.
\end{aligned}$$

Under the assumption that Y_{ij} are normally distributed, the three terms in the last row are independent random variables. Moreover, the first term, upon division by σ^2 , has the chi-square distribution with the number of degrees of freedom equal $(n_1 - 1) + \cdots + (n_I - 1) = N - I$ regardless of whether or not the null hypothesis is true. The second sum, again upon division by σ^2 , has the chi-square distribution with $I - 1$ degrees of freedom, provided that $E(\bar{Y}_{i+} - \bar{Y}_{++}) = \alpha_i$ for every i . Thus, under H_0 , the sum

$$\frac{1}{\sigma^2} \sum_{i=1}^I n_i (\bar{Y}_{i+} - \bar{Y}_{++})^2 \quad (13.25)$$

has a chi-square distribution with $I - 1$ degrees of freedom, and any violation of H_0 will increase the expectation of the sum (13.25). This suggests the use of an appropriate F ratio to test the hypothesis H_0 .

Finally, if $\mu = 0$, then $N(\bar{Y}_{++})^2/\sigma^2$ has the chi-square distribution with one degree of freedom, which allows us to test hypothesis $H'_0 : \mu = 0$ against the alternative $H'_1 : \mu \neq 0$. In terms of observations, the testing procedure is most often displayed in the form of the following ANOVA table:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F ratio
Mean	1	SSM	$MSM = \frac{SSM}{1}$	$F_M = \frac{MSM}{MSR}$
Factor	$I - 1$	SSA	$MSA = \frac{SSA}{I-1}$	$F_A = \frac{MSA}{MSR}$
Residual	$N - I$	SSR	$MSR = \frac{SSR}{N-I}$	
Total	N	SST		

where

$$\begin{aligned}
SSM &= N(\bar{y}_{++})^2, & SSA &= \sum_{i=1}^I n_i (\bar{y}_{i+} - \bar{y}_{++})^2, \\
SSR &= \sum_{i=1}^I \sum_{j=1}^{n_i} n_i (y_{ij} - \bar{y}_{i+})^2, & SST &= \sum_{i=1}^I \sum_{j=1}^{n_i} y_{ij}^2.
\end{aligned}$$

We reject the hypothesis $H_0 : \alpha_1 = \cdots = \alpha_I = 0$ (at level γ) if the ratio F_A exceeds the upper quantile $F_{\gamma, I-1, N-I}$. It is important to remember that in ANOVA *one always uses a one-sided critical region* (since any violation of H_0 tends to increase the numerator without affecting the denominator).

If, for some reason, one is interested in the hypothesis $H'_0 : \mu = 0$, then one should reject it in favor of $H'_1 : \mu \neq 0$ if the ratio F_M exceeds $F_{\gamma, 1, N-I}$. Such a test is useful if observations can be positive as well as negative (e.g., deviations from the required standards in a technological process).

PROBLEMS

13.11.1 Verify that the cross-products in (13.24) do indeed equal zero.

13.11.2 To test the mileage achieved by cars produced by different companies, but of comparable price, size, and so on, one make of cars was selected from among the three major American companies and two foreign companies. For each make selected, a number of new cars was chosen and their mpg (miles per gallon) recorded. The data (in mpg) are as follows:

$n_1 = 5$	$n_2 = 4$	$n_3 = 5$	$n_4 = 3$	$n_5 = 6$
25.1	27.1	39.9	25.4	29.2
26.2	26.4	21.4	28.2	29.3
24.9	26.8	22.2	27.1	30.4
25.3	27.2	22.5		28.5
23.9		20.8		28.9
				29.2

At the significance level 0.05, test the hypothesis that the average mpg is the same for all makes of cars tested.

13.11.3 Suppose that we take a random sample of size n from a normal distribution $N(\mu, \sigma^2)$. We divide the observations into k groups of sizes $n_1, \dots, n_k$, where $n_i \geq 2$ for $i = 1, \dots, k$ and $n_1 + \dots + n_k = n$. Let S_i^2 be the sample variance in the i th group. Find: **(i)** The distribution of $[(n_1 - 1)S_1^2 + \dots + (n_k - 1)S_k^2]/\sigma^2$. **(ii)** The distribution of S_k^2/S_1^2 .

13.11.4 Show that the test developed in this chapter is equivalent to the Student's t test for the case when the factor operates at two levels only.

13.12 TWO-WAY LAYOUT

Assume now that we have data concerning possible effects of two factors, A and B , with I and J levels, respectively. Let y_{ij} be the observation for the i th level of A and the j th level of B . Such data can be arranged into a matrix $[y_{ij}]$ with I rows and J columns. For the moment assume that we have one observation in each of the IJ cells formed by crossing factors A and B . As before, we regard y_{ij} as the recorded value of a random variable Y_{ij} , assumed to have normal distribution $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$, where α_i and β_j represent the effects of factors A and B . Without loss of generality, we can write

$$\sum_{i=1}^I \alpha_i = \sum_{j=1}^J \beta_j = 0. \quad (13.26)$$

We want to construct a test for the hypothesis

$$H_0^{(A)} : \alpha_1 = \dots = \alpha_I = 0 \quad \text{against} \quad H_1^{(A)} : \alpha_i \neq 0 \text{ for some } i,$$

as well as a test for

$$H_0^{(B)} : \beta_1 = \cdots = \beta_J = 0 \quad \text{against} \quad H_1^{(B)} : \beta_j \neq 0 \text{ for some } j.$$

The tests are built on a partition of the sum of squares, as in the case of one factor. Omitting again the cross-products (which are zero), we have

$$\begin{aligned} S^2 &= \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \mu - \alpha_i - \beta_j)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{++} + \bar{Y}_{++} - \mu - \alpha_i - \beta_j)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{++} - \alpha_i - \beta_j)^2 + IJ(\bar{Y}_{++} - \mu)^2 \\ &= \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} - \bar{Y}_{i+} - \bar{Y}_{+j} + \bar{Y}_{++})^2 + \sum_{i=1}^I J(\bar{Y}_{i+} - \bar{Y}_{++} - \alpha_i)^2 \\ &\quad + \sum_{j=1}^J I(Y_{+j} - \bar{Y}_{++} - \beta_j)^2 + IJ(\bar{Y}_{++} - \mu)^2. \end{aligned}$$

Substituting the observed data for random variables and letting $\mu = 0, \alpha_i = \beta_j = 0$ for all i, j , we obtain the following sums of squares:

$$\begin{aligned} SSM &= IJ(\bar{y}_{++})^2, & SSA &= \sum_{i=1}^I J(\bar{y}_{i+} - \bar{y}_{++})^2, \\ SSB &= \sum_{i=1}^I I(\bar{y}_{+j} - \bar{y}_{++})^2, & SSR &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{i+} - \bar{y}_{+j} + \bar{y}_{++})^2, \end{aligned}$$

which add up to the total sum of squares

$$SST = \sum_{i=1}^I \sum_{j=1}^J y_{ij}^2.$$

The four terms in the last decomposition of S^2 are independent random variables. The SSR does not involve any parameters and (upon division by σ^2) has a chi-square distribution with the number of degrees of freedom equal to

$$IJ - (I - 1) - (J - 1) - 1 = (I - 1)(J - 1),$$

regardless of whether the hypotheses are true or false. The SSA sum has a chi-square distribution if $\alpha_1 = \cdots = \alpha_I = 0$, and a similar statement holds for SSB if $\beta_1 = \cdots = \beta_J = 0$. The numbers of degrees of freedom are $I - 1$ and $J - 1$, respectively. Any deviation from the null hypothesis tends to increase the corresponding sum of squares.

The ANOVA table is now:

Source of variation	Degrees of freedom	Sum of squares	Mean sum of squares	F ratio
Mean	1	SSM	$MSM = \frac{SSM}{1}$	$F_M = \frac{MSM}{MSR}$
A	$I - 1$	SSA	$MSA = \frac{SSA}{I-1}$	$F_A = \frac{MSA}{MSR}$
B	$J - 1$	SSB	$MSB = \frac{SSB}{J-1}$	$F_B = \frac{MSB}{MSR}$
Residual	$(I - 1)(J - 1)$	SSR	$MSR = \frac{SSR}{(I-1)(J-1)}$	
Total	$N = IJ$	SST		

The hypothesis $H_0^{(A)}$ (that factor A has no effect) is rejected whenever $F_A > F_{\gamma, I-1, (I-1)(J-1)}$, and $H_0^{(B)}$ is rejected if $F_B > F_{\gamma, J-1, (I-1)(J-1)}$, where γ is the desired level of significance.

PROBLEMS

13.12.1 Twelve subjects participated in a study comparing the effectiveness of three weight-reducing diets. The subjects were grouped according to their initial weight, and each of three subjects from each initial weight group was randomly assigned to a diet. The weight loss (in pounds) at the end of the experimental period is given below:

Initial Weight	Diet		
	A	B	C
150–174	10	23	24
175–199	12	21	26
200–224	12	31	21
Over 224	20	28	33

(i) Do these data provide sufficient evidence that (after eliminating the effect of initial weight) the diets are different in their effectiveness? Use $\alpha = 0.01$. (ii) Does the initial weight affect the loss of weight?

13.12.2 The nutritional value of a certain vegetable is measured on 18 specimens grown in two varieties in three geographical regions. The data are as follows:

Variety	Geographical Region		
	A	B	C
1	6.3	9.2	6.8
	11.5	5.1	7.2
	9.2	8.1	5.5
	11.0	5.4	7.1
2	7.3	5.0	7.8
	8.2	6.1	8.4

Study the effect of variety and geographical location on the nutritional value, taking the average of three observations in each cell as a response. Use the significance level $\alpha = 0.05$.

- 13.12.3** The model for three factors is $E(Y_{ijk}) = \mu + \alpha_i + \beta_j + \gamma_k$, where $\sum \alpha_i = \sum \beta_j = \sum \gamma_k = 0$. Assume that Y_{ijk} is normally distributed with variance σ^2 . Derive the tests for the hypotheses analogous to those in the case of two factors.

13.13 ANOVA MODELS WITH INTERACTION

The model considered in Section 13.12 was of the form $E(Y_{ij}) = \mu + \alpha_i + \beta_j$, by which we assume that the effects of the factors A and B are additive. The hypotheses $H_0^{(A)}$ and $H_0^{(B)}$ are tested within this model.

In general, the effects of A and B need not be additive, and one might wish to consider the case where $E(Y_{ij})$ is an entirely arbitrary function of i and j . Such a function can always be represented in the form

$$E(Y_{ij}) = \mu + \alpha_i + \beta_j + \gamma_{ij},$$

where $\sum_i \alpha_i = \sum_j \beta_j = \sum_{ij} \gamma_{ij} = 0$ for all i, j (we leave the proof as an exercise). The constants γ_{ij} are referred to as interaction terms. Testing for the presence of interaction is based on an idea similar to that used in testing linearity of regression. It requires a model-independent estimate of σ^2 , which in turn can be achieved if we have more than one observation for each combination of levels of factors A and B . We assume therefore that the data now have the form of a three-dimensional array y_{ijk} , $i = 1, \dots, I; j = 1, \dots, J; k = 1, \dots, K$, where $I \geq 2, J \geq 2, K \geq 2$. Leaving the details of the derivation as an exercise, we can write the decomposition of the sum of squared deviations of the random variables Y_{ijk} from their means as

$$\begin{aligned} S^2 &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (Y_{ijk} - \mu - \alpha_i - \beta_j - \gamma_{ij})^2 \\ &= IJK(\bar{Y}_{+++} - \mu)^2 \\ &\quad + \sum_{i=1}^I JK(\bar{Y}_{i++} - \bar{Y}_{+++} - \alpha_i)^2 + \sum_{j=1}^J IK(\bar{Y}_{+j+} - \bar{Y}_{+++} - \beta_j)^2 \\ &\quad + \sum_{i=1}^I \sum_{j=1}^J K(\bar{Y}_{ij+} - \bar{Y}_{i++} - \bar{Y}_{+j+} + \bar{Y}_{+++} - \gamma_{ij})^2 \\ &\quad + \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (Y_{ijk} - \bar{Y}_{ij+})^2. \end{aligned} \tag{13.27}$$

As before, the last sum (upon division by σ^2) has the chi-square distribution with $IJ(K - 1)$ degrees of freedom, given only normality and homoscedasticity (regardless of any other hypotheses tested). Thus, it can serve as a denominator in all F ratios used for testing, while the two single sums and the double sum can be used to test the hypotheses about the effects of A , of B , and of their interaction. To put it differently, each of the sums above, divided by its number of degrees of freedom, is an unbiased estimator of σ^2 if the appropriate hypothesis

is true (except for the last sum, for which this holds regardless of any hypothesis). Letting

$$\begin{aligned}
 SSM &= IJK(\bar{y}_{+++})^2, \\
 SSA &= \sum_{i=1}^I JK(\bar{y}_{i++} - \bar{y}_{+++})^2, \\
 SSB &= \sum_{j=1}^J IK(\bar{y}_{+j+} - \bar{y}_{+++})^2, \\
 SSAB &= \sum_{i=1}^I \sum_{j=1}^J K(\bar{y}_{ij+} - \bar{y}_{i++} - \bar{y}_{+j+} + \bar{y}_{+++})^2, \\
 SSR &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \bar{y}_{ij+})^2, \\
 SST &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K y_{ijk}^2,
 \end{aligned}$$

the corresponding ANOVA table is as follows:

Source of Variation	Degrees of Freedom	Sum of Squares	Mean Sum of Squares	F Ratio
Mean	1	SSM	$MSM = \frac{SSM}{1}$	$F_M = \frac{MSM}{MSR}$
A	$I - 1$	SSA	$MSA = \frac{SSA}{I-1}$	$F_A = \frac{MSA}{MSR}$
B	$J - 1$	SSB	$MSB = \frac{SSB}{J-1}$	$F_B = \frac{MSB}{MSR}$
AB	$(I - 1)(J - 1)$	$SSAB$	$MSAB = \frac{SSI}{(I-1)(J-1)}$	$F_{AB} = \frac{MSI}{MSR}$
Residual	$IJ(K - 1)$	SSR	$MSR = \frac{SSR}{IJ(K-1)}$	
Total	$N = IJK$	SST		

■ EXAMPLE 13.16

A researcher studies the effects of sex and type of stimulus (“soothing” or “exciting”) on the aggressive behavior of parrots. Six male and six female birds of a given species are each placed in a separate cage isolated from other cages. The six birds of a given gender are randomly divided into two groups. The cages are covered for the night. Before uncovering the cage in the morning, the birds hear a tape. One is a “soothing” tape, with the voice of the experimenter talking quietly to the birds. The other tape contains angry voice of the experimenter. The observed value Y is the number of times the bird attacks the experimenter’s hand when she uncovers the cage and puts food into

the plate. The data are as follows (the three numbers in each cell represent the data for three birds):

	Soothing (S)	Angry (A)
M	8 6 13	22 28 33
F	5 10 6	12 14 9

The within-cell averages $\bar{y}_{ij+}$ and row and column averages Y_{i+} , y_{+j} , as well as y_{++} are

	S	A	
M	9	27.67	18.33
F	7	11.67	9.33
	8	19.67	13.83

The SSR of the squared deviations from cell means is $SSR = 1, 113.33$. We therefore have the next table:

	df	SS	MS	F
Mean	1	2,296.33	2,296.33	162.10
Gender	1	243.00	243.00	17.15
Stimulus	1	408.33	408.33	28.82
Interaction	1	147.00	147.00	10.38
Residuals	8	113.33	14.17	

Since $F_{0.05,1,8} = 5.318$, we can conclude that both the gender of the parrot and the type of stimulus have an effect, and there is interaction between gender and type of stimulus at the 0.05 significance level. The mean response is significantly different than zero.

To better grasp the meaning of interaction, we will use a graphical representation. Let us arrange the categories of one factor, say A , along the horizontal axis (there may be no numerical values attached, and the categories need not have any “natural” order). Then along the horizontal axis, we can plot average responses $\bar{y}_{ij+}$ for various j . The effects of A , B and of their interaction can now be interpreted as follows (see Figure 13.4).

The effect of A means that at least one of the curves differs significantly from the horizontal line (last two figures). The effect of B , but not of any interaction, means that the curves for various levels of B are parallel to one another but significantly different from one another (Figure 13.4b and c). Finally, the interaction reveals itself by the lack of parallelism of the curves (Figure 13.4d).

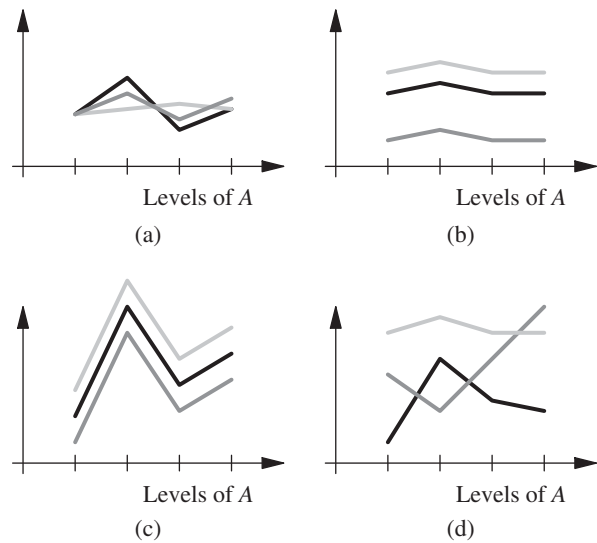


Figure 13.4 (a) No effects of A , B or AB . (b) No effect of A , effect of B , no interaction. (c) Effect of A , no effect of B , no interaction. (d) Effect of A , effect of B , effect of interaction.

PROBLEMS

13.13.1 The scores measuring the level of emotional maturity of young adult males classified by age and extent of use of marijuana are given below.

Age	Marijuana use		
	Never	occasionally	Daily
15–19	25	18	17
	28	23	24
	22	19	19
20–24	28	16	18
	32	24	22
	30	20	20
25–29	25	14	10
	35	16	8
	30	15	12

Test for effects of age, extent of use, and their interaction. Use $\alpha = 0.05$.

13.13.2 Use data given in Problem 13.12.2 to test for the existence of interaction between variety and geographical notation.

13.13.3 Show that if $E(Y_{ij}) = c(i, j) = \mu + \alpha_i + \beta_j + \gamma_{ij}$, where $i = 1, \dots, I, j = 1, \dots, J$, then one can always find constants μ, α_i, β_j , and γ_{ij} such that

$$\sum_{i=1}^I \alpha_i = \sum_{j=1}^J \beta_j = \sum_{i=1}^I \sum_{j=1}^J \gamma_{ij} = 0.$$

13.14 FURTHER EXTENSIONS

The ANOVA models presented in the last three sections can easily be extended to the case of more than two factors, with or without interaction, provided that we consider completely balanced designs. Each level of a factor can be combined with all combinations of levels of other factors, and in each cell we have the same number of observations. Testing for interactions is possible only if the number of observations per cell is at least two. The number of observations needed very quickly becomes unattainable practically. For instance, with three factors, each on five levels, we need $5^3 = 125$ observations, and twice that if we want to test for interactions.

This situation led to research in two major directions. The first was to invent experimental schemes that allow testing for the presence of effects (as well as estimation of those effects) with as small a number of experiments as possible. If one resigns from the stringent requirement that every level of a factor has to appear with every combination of levels of other factors, there are many possibilities of experiments (e.g., forming the so-called Latin or Greco-Latin squares). To use a simple example, imagine that we have three factors, each appearing on five levels. Representing one factor as a row, the other as a column, and the third as letter (with levels a, b, c, d, e), we can arrange the experiment as follows:

a	b	c	d	e
b	c	d	e	a
c	d	e	a	b
d	e	a	b	c
e	a	b	c	d

Each level of the first factor (row) combines exactly once with each level of the second factor (column), and exactly once with the third factor (letters). The same is true for the other two factors. However, of $5^3 = 125$ possible combinations, only 25 actually appear.

It is possible to include here the fourth factor (e.g., Greek letters, $\alpha, \beta, \gamma, \delta, \epsilon$) as follows:

$a\alpha$	$b\beta$	$c\gamma$	$d\delta$	$e\epsilon$
$b\epsilon$	$c\alpha$	$d\beta$	$e\gamma$	$a\delta$
$c\delta$	$d\epsilon$	$e\alpha$	$a\beta$	$b\gamma$
$d\gamma$	$e\delta$	$a\epsilon$	$b\alpha$	$c\beta$
$e\beta$	$a\gamma$	$b\delta$	$c\epsilon$	$d\alpha$

Now each row and each column has exactly one of the Roman letters and exactly one of the Greek letters, and each Roman letter is combined exactly once with each Greek letter.

One can therefore plan an agricultural experiment (say) in which in each row are plants of one of five varieties of seed, and in each column one of the five varieties of fertilizers. Then, each Roman letter would correspond to one of the five different amounts of watering, and each Greek letter to one of the five different times of planting.

There are seemingly countless variety of experimental designs to cover all contingencies that occur in practice, each design with its own testing or estimation procedures. By introducing appropriate criteria, one can search for designs that optimize these criteria.